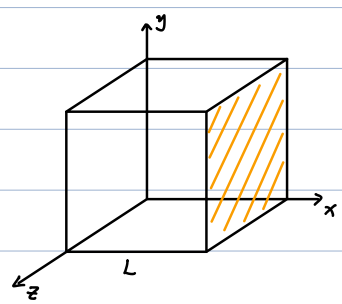
Kinetic theory of gases

Ideal gas

- Negligible molecular volume (compared to distance)
- Short-range force (active only during collision)
- Elastic collisions
- Identical molecules



Collision on the RHS wall: $v_x \to -v_x$

$$\Delta p_x = (-m v_x) - m v_x = -2 m v_x$$

time interval between collisions:

$$\Delta t = \frac{2L}{v_x}$$

Force from a single molecule:

$$F_x = \frac{|\Delta p_x|}{\Delta t} = \frac{2 m v_x}{2L/v_x} = \frac{m v_x^2}{L}$$

Pressure:

$$P = \frac{F_{x1} + F_{x2} + \cdots + F_{xN}}{L^2}$$

$$= \frac{m}{L^3}(v_{x1}^2 + v_{x2}^2 + \cdots + v_{xN}^2)$$

$$= \frac{Nm}{L^3}\overline{v_x^2} \qquad \overline{v_x^2} \equiv \frac{1}{N}(v_{x1}^2 + v_{x2}^2 + \cdots v_{xN}^2)$$

$$= \frac{n N_A m}{L^3}\overline{v_x^2} \qquad n = \frac{N}{N_A}$$

$$\begin{cases} V = L^3 \\ v^2 = v_x^2 + v_y^2 + v_z^2 \end{cases}$$

$$\Rightarrow \quad P = \frac{n N_A m}{3V}\overline{v^2}$$

$$\Rightarrow \quad nRT = \frac{n N_A m}{3}\overline{v^2}$$

$$\Rightarrow \quad v_{rms} \equiv \sqrt{\overline{v^2}} = \sqrt{\frac{3RT}{N_A m}}$$

$$= \sqrt{\frac{3k_B T}{m}}$$

rms: root-mean-square (方均根)

Average translational kinetic energy per molecule:

$$K_{avg} = \frac{1}{2} m \overline{v^2} = \frac{1}{2} m v_{rms}^2$$

Since $v_{rms} = \sqrt{\frac{3k_B T}{m}}$:

$$\boxed{K_{avg} = \frac{3}{2} k_B T}$$

- Temperature is a measure of the molecular kinetic energy

Equipartition theorem (more serious proof later): 能量均分定理

Average kinetic energy per degree of freedom is $\begin{cases} \frac{1}{2} k_B T & \text{per molecule} \\ \frac{1}{2} RT & \text{per mole} \end{cases}$

e.g. 3 translational modes:

$$\frac{1}{2} m \overline{v_x^2} = \frac{1}{2} m \overline{v_y^2} = \frac{1}{2} m \overline{v_z^2} = \frac{1}{2} k_B T$$

Internal energy (monoatomic gas)

$$U = \frac{3}{2} N k_B T$$
$$= \frac{3}{2} n R T$$

$R = N_A k_B$

Heat capacity (热容):

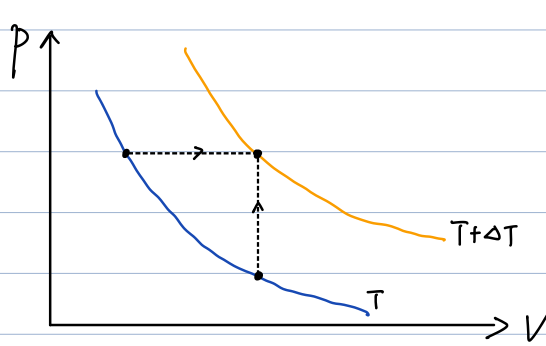
$$C \equiv \frac{\Delta Q}{\Delta T}$$

Specific heat capacity, abbr "specific heat" (比热、比热容):

$$c \equiv \frac{C}{M}$$ (heat capacity per mass)

Molar heat capacity (摩尔热容):

$$c \equiv \frac{C}{n}$$ (heat capacity per mole)



Molar heat capacity at constant V (monoatomic gas) :

$$c_v = \frac{C}{n} = \frac{1}{n}\left(\frac{\Delta Q}{\Delta T}\right)_v$$ 定容摩尔热容

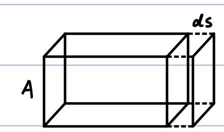
$$= \frac{1}{n}\left(\frac{dU}{dT}\right)_v$$

$$= \frac{3}{2}R$$

Molar heat capacity at constant p (monoatomic gas) :

$$c_p = \frac{C}{n} = \frac{1}{n}\left(\frac{\Delta Q}{\Delta T}\right)_p$$ 定压摩尔热容

$$= \frac{1}{n}\left(\frac{\Delta U + W}{\Delta T}\right)_p$$



$$W = F\,ds = pA\,ds = p\,dV$$

$$\Rightarrow c_p = \frac{1}{n}\left(\frac{dU}{dT} + p\frac{dV}{dT}\right)_p$$ $$pV = nRT \Rightarrow p\left(\frac{dV}{dT}\right)_p = nR$$

$$\Rightarrow \boxed{c_p = c_v + R}$$

Diatomic / Polyatomic gas

e.g. $\begin{cases} He : \text{monoatomic} \\ H_2 : \text{diatomic} \\ CO_2 : \text{polyatomic} \end{cases}$

$$\begin{cases} U = \frac{f}{2}Nk_BT = \frac{f}{2}nRT \\ c_v = \frac{f}{2}R \\ c_p = c_v + R \end{cases}$$ > SAME at different T !

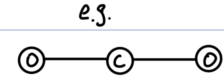
Degrees of freedom :

$$f = \underbrace{t}_{\text{translation}} + \underbrace{r}_{\text{rotation}} + \underbrace{2s}_{\text{oscillation}}$$

N-atom linear molecule :

$$\begin{cases} t = 3 \\ r = 2 \\ s = 3N - 5 \end{cases}$$

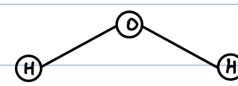
e.g.



N-atom nonlinear molecule :

$$\begin{cases} t = 3 \\ r = 3 \\ s = 3N - 6 \end{cases}$$

e.g.



Q : Why 2s ?

oscillation $\begin{cases} \text{kinetic energy} & \frac{1}{2}RT \\ \text{potential energy} & \frac{1}{2}RT \end{cases}$

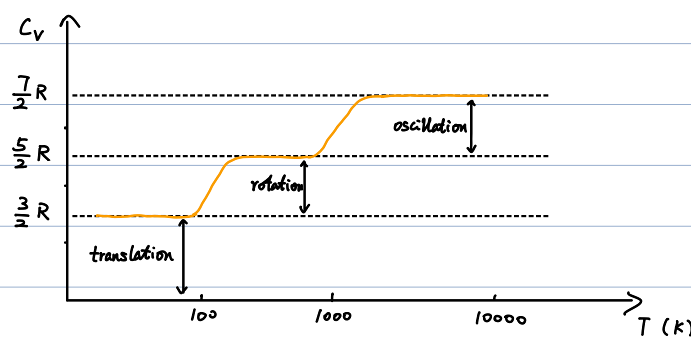
Example. H_2

$t = 3, \quad r = 2, \quad s = 3N - 5 = 1$

$\Rightarrow f = t + r + 2s = 7$

$\Rightarrow C_V = \frac{7}{2}R$

Reality :



(origin of discrepancy : quantum mechanics)

Statistics

Example.

Speeds of 10 particles : $v_1 = 0, \; v_2 = 1, \; \cdots \; v_{10} = 9$ m/s

Average velocity : $\bar{v} = \frac{1}{N}\sum_i v_i = \frac{1}{10}(0 + 1 + \cdots + 9) = 4.5$ m/s

Root-mean-square velocity : $v_{rms} = \sqrt{\frac{1}{N}\sum_i v_i^2} = \sqrt{\frac{1}{10}(0^2 + 1^2 + \cdots + 9^2)} \approx 5.34$ m/s

Probability distribution function $f(v)$:

Probability to find particle's velocity in $[v, v+dv]$ is $f(v)dv$.

Normalization:

$$\int_{-\infty}^{\infty} f(v)\,dv = 1$$

Average velocity:

$$\bar{v} = \int_{-\infty}^{\infty} v\, f(v)\,dv$$

Root-mean-square velocity:

$$v_{rms} = \sqrt{\int_{-\infty}^{\infty} v^2 f(v)\,dv}$$

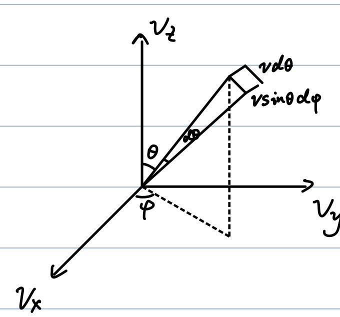
Higher dimension:

$$f(v)dv \longrightarrow f(\vec{v})\, dv_x\, dv_y\, dv_z$$

Maxwell distribution of velocity (速度):

$$\boxed{\begin{aligned} f(\vec{v}) &= f(v_x)f(v_y)f(v_z) \\ &= \left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-\frac{mv^2}{2k_BT}} \end{aligned}}$$

$v \equiv |\vec{v}|$



Probability to find $\vec{v}$ in $[v, v+dv] \times [\theta, \theta+d\theta] \times [\varphi, \varphi+d\varphi]$:

$$f(\vec{v})\, v^2 \sin\theta\, dv\, d\theta\, d\varphi$$

Isotropic (no dependence of $f(\vec{v})$ on θ, φ)

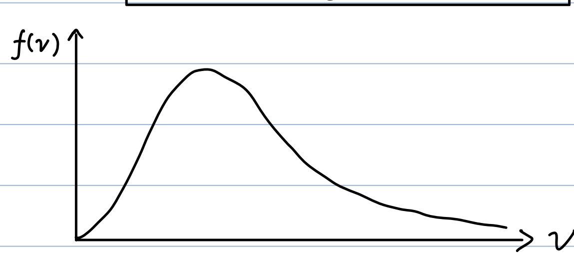
$\Rightarrow$ Probability to find $\vec{v}$ in $[v, v+dv]$: (any θ, φ)

$$\int_0^{\pi} d\theta \int_0^{2\pi} d\varphi\, f(\vec{v})\, v^2 \sin\theta\, dv$$

$$= 4\pi\, f(\vec{v})\, v^2\, dv$$

Maxwell distribution of speed (速率):

$$\boxed{f(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-\frac{mv^2}{2k_BT}} \cdot v^2}$$



Characteristic speeds

Most probable speed v_p （最概然速率）

$$\frac{df(v)}{dv} = 0$$

$$\Rightarrow \frac{d}{dv}\left(e^{-\frac{mv^2}{2k_BT}} \cdot v^2\right) = 0$$

$$\Rightarrow e^{-\frac{mv^2}{2k_BT}} \cdot \left(-\frac{mv}{k_BT}\right) v^2 + e^{-\frac{mv^2}{2k_BT}} \cdot 2v = 0$$

$$\Rightarrow \boxed{v_p = \sqrt{\frac{2k_BT}{m}}}$$

Average speed $\bar{v}$

$$\bar{v} = \int_0^\infty dv\, f(v)\, v$$

$$= 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} \int_0^\infty dv\, e^{-\frac{mv^2}{2k_BT}} \cdot v^3$$

$$\Rightarrow \boxed{\bar{v} = \sqrt{\frac{8k_BT}{\pi m}}} \approx \sqrt{\frac{2.55\, k_BT}{m}}$$

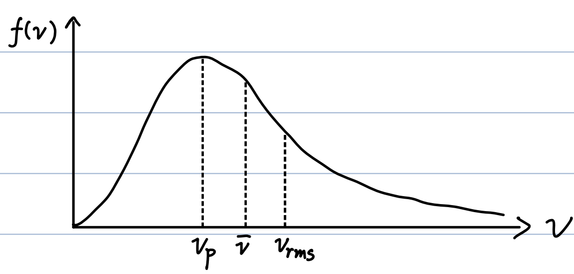
Root-mean-square speed v_{rms}

$$v_{rms}^2 = \int_0^\infty dv\, f(v)\, v^2$$

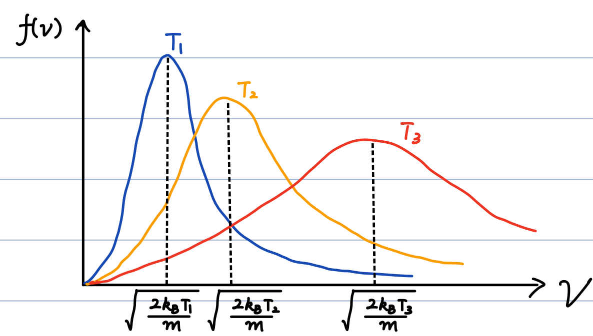
$$= 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} \int_0^\infty dv\, e^{-\frac{mv^2}{2k_BT}} \cdot v^4$$

$$= \frac{3k_BT}{m}$$

$$\Rightarrow \boxed{v_{rms} = \sqrt{\frac{3k_BT}{m}}}$$



Varying temperature



$T_1 < T_2 < T_3$

Maxwell – Boltzmann distribution (abbr "Boltzmann distribution")

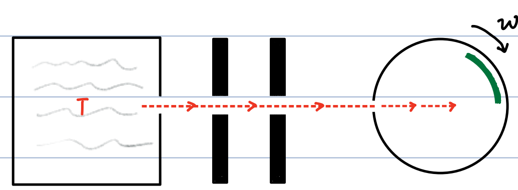
$$f_{MB}(\vec{r}, \vec{v}) = f(x,y,z)\, f(v_x, v_y, v_z)$$

$$= \frac{1}{\int e^{-\varepsilon_p(x,y,z)/k_BT}\, dx\,dy\,dz}\, e^{-\varepsilon_p(x,y,z)/k_BT} \cdot \left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-\frac{mv^2}{2k_BT}}$$

$$\propto \boxed{e^{-[\frac{1}{2}mv^2 + \varepsilon_p(\vec{r})]/(k_BT)}}$$

$\varepsilon_p(\vec{r})$: potential energy at $\vec{r}$.

Maxwell distribution – experimental verification



particles with different v hits different part of the target.